PREVENTION OF CHILDHOOD BLINDNESS

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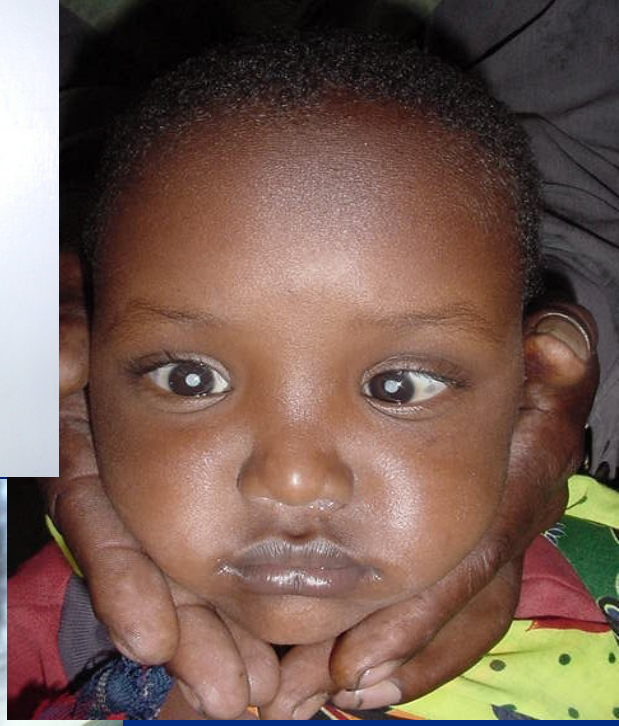


INTRODUCTION: A CHILD;

- Is a precious resource of a family,
- Represents the family's future and hopes
- A child whose blindness could have been prevented or cured is a great disaster

Childhood blindness- WHO Definitions

Childhood: 0-15 YRS (WHO)
0-16 YRS (UNICEF)

- Blindness: BCVA <3/60 in the better eye or central field <10°
- *Blindness from any cause occurring in a person aged 15 yrs or less*

Childhood blindness

- *About 1.5 million children are blind worldwide(* 20% live in Sub-Saharan Africa & 50% in Asia
- *Occurs in 5 - 10/10,000 children Globally*
- *About 500,000 children become blind every year*
- *One Child becomes blind every minute*
- *Half (50%) of them die within one or two years of becoming blind*

Childhood blindness Prevention? WHY a priority

- Large number of “blind person years” (75 million blind person yrs)
- Causes are many & different from those of adults
- Most of the causes are avoidable (70 to 80%)

- Blinding conditions are also causes of child mortality
- Examination and assessment is challenging

- Requires specific management expertise
- Irreversible amblyopia if treatment is delayed

Childhood Blindness – V2020 Global Initiative

Goal:

To reduce childhood blindness from 10/10,000 Children to 4/10,000 Children by the year 2020

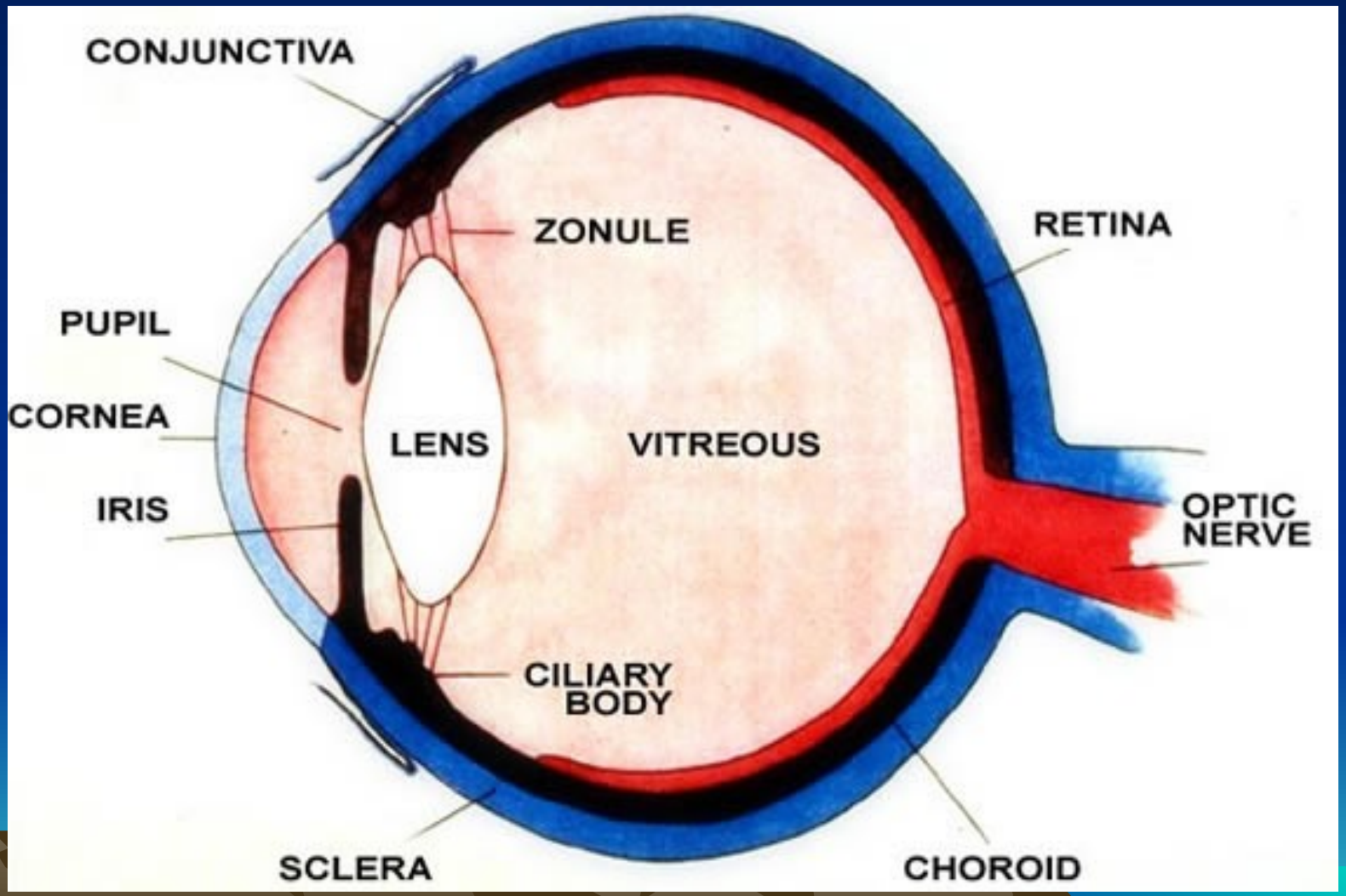
Key elements

-Strategies for disease control

-Human resource need & development

-Infrastructure/technology needs & development

THE EYE


CONJUNCTIVA
ZONULE
RETINA
PUPIL
CORNEA
LENS
VITREOUS
IRIS
OPTIC NERVE
CILIARY BODY
SCLERA
CHOROID

Visual Development

- At birth vision is poor, but develops rapidly with optimal/undisturbed visual inputs
- Blink reflex to bright light –days after birth
- 6/52 – mother/ baby eye contact
- 2-3/12 – interested with bright objects
- development & maturation of visual function occurs from birth & complete by 9 years.

Signs of poor visual development

- Wandering eye movements
- Lack of response to familiar object & faces
- Staring at bright lights
- Forceful rubbing of eyes
- Nystagmus or “shaking eyes”
- Sunken eye appearance

Major causes of childhood blindness

1. **Corneal blindness** due to:-

a) VAD -Nutritional & Measles

b) HTEM

c) Corneal ulcers

d) Neonatal conjunctivitis

Causes of Childhood blindness ct

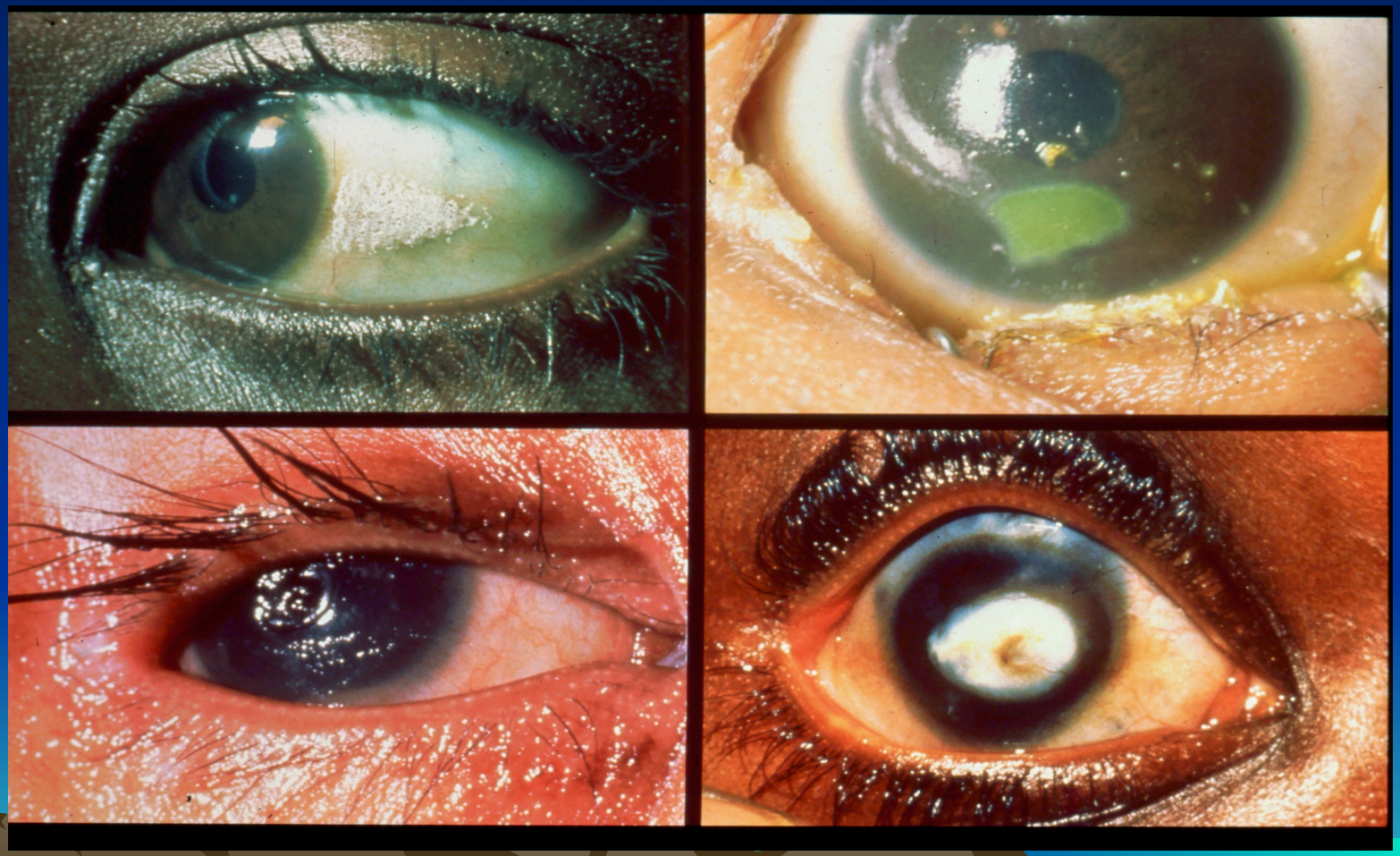
2. Congenital/paediatric glaucomas
3. Paediatric Cataracts
4. Retinoblastoma
5. Retinopathy Of Prematurity (ROP)
6. Uncorrected high Refractive Errors
7. Squints & Amblyopia
8. Cerebral visual impairment

1.CORNEAL SCARS –VAD

why children are at risk (6 months to 6 yrs)

- Vitamin A Liver stores relatively low at birth
- Breast milk supplies adequate retinol for 6 months
- High requirement because of growth
- High rate of infection and diarrhoea after weaning
- Measles infection after the age of 6 months
- Totally dependant on adults

Clinical presentation of VAD



Prevention of VAD

- Breast feeding
- Prophylactic vitamin A every 6/12
- Supplement feeding of infants with green yellow leafy vegetables & proteins
- Measles immunization
- avoid use of HTEM
- Food fortification
- Family planning

CORNEAL SCARS:
2. Neonatal conjunctivitis

Prevention of Neonatal Conjunctivitis

- Early identification & treatment of mothers in 3rd trimester of pregnancy
- Antenatal screening of high risk women
- Ocular prophylaxis
- Ensure aseptic delivery
- Early detection, referral & appropriate treatment of newborn conjunctivitis

3. CONGENITAL GLAUCOMA

Childhood glaucoma can affect one or both eyes and can

-present at birth [congenital]

-develop during childhood

NB-Before age 3 yrs a child's eye is capable of stretching if eye pressure rises BUPHTHALMOS or OX eye

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Symphtoms & signs

- Excessive tearing
- Photophobia
- Blepharospasms

- Clouding of cornea +/- increase in cornea & eyeball size
- Red eye, poor vision in older children

How should we treat GLAUCOMA

- Early detection & referal
- Glaucoma surgery
- Longterm follow-up to monitor:

  Vision, IOP, Cornea size & clarity, refraction, Fundoscopy, surgical site
- Visual Rehabilitation

4. CONGENITAL CATARACT

Causes: -Congenital- INFECTIONS
-HEREDITARY
-METABOLIC
-50% idiopathic
-Trauma

Symptoms:

Poor vision, Can affect one or both eyes

Signs:- white pupil reflex (LEUCOKORIA)
-May have other abnormalities

MANAGEMENT OF CONGENITAL CATARACT

- Immunization --rubella
- Early detection
- early referral
- Early Surgery
- Early visual rehabilitation

5. RETINOBLASTOMA –cancer of the eye

Causes: may be inherited or sporadic

- Can affect one or both eyes and children below 5 years
- Signs and symptoms: white pupil reflex (LEUCOKORIA), poor vision, squint.
- If not treated early it may grow very big and even cause death
- Action: Refer IMMEDIATELY

RETINOBLASTOMA

D/DX OF LEUCOKORIA

- Cataract
- Retinoblastoma
- Retina detachment from any cause
- Persistent hypo plastic primary vitreous (PHPV)
- Retinopathy of prematurity
- Coloboma of the choroid
- Toxocara endophthalmitis/granulomas
- Endophthalmitis –intra ocular infection (bacteria, fungi)
- Coats disease

6. Refractive errors

Refractive error

- Definition: poor vision due to defect in focusing mechanism of the eye
- which can be corrected with spectacles or contact lenses

- Types: Myopia, Hypermetropia, astigmatism, presbyopia
- Signs and symptoms: Poor vision, Itchy, teary eyes, headache, Unusual behaviour- “Squinting”.

Refractive errors

- Treatment - Child will benefit from spectacles

- If not treated early

  -Poor performance.

  - May develop “lazy eyes” -Amblyopia

7. SQUINTS & AMBLYOPIA

Causes & Mx of SQUINTS

- Defn : misaligned eyes

Causes:

- Sensory factors - Any form of eye disease which interferes with vision
- Motor factors – Muscles & nerves
- Central factors

NB: refer all children presenting with squint for further evaluation & management

8. CEREBRAL VISUAL IMPAIRMENT

Defn: Visual impairment involving the posterior visual pathway and can be associated with cerebral palsy

Causes; birth asphyxia, trauma, brain infections, hydrocephalus, epilepsy, etc

Prevention: good management of labour &

early ,appropriate mx of childhood infections

8. Retinopathy Of Prematurity

- Retinopathy in preterm infants< 1500gm or born before 34 wks gestation
- Child becomes blind before 6 months of age (3.5-18% blindness) from retina detachment
- Severe, progressive visual impairment in others
- The child remains blind the rest of their life.
- Effective and safe prophylactic treatment is available (laser Px)

There is no DISEASE AT BIRTH !!

Prevention of ROP – Early screening of Preterm babies

Whom to screen-

- **Any baby you call "preterm" should be screened, irrespective of other criteria**
- **Birth weight: < 1700 gms**
- **Gestational age < 34-35 weeks When to Screen: "Day -- 30 strategy"**

One eye examination should be definitely over before day 30 of life of the premature infant. Day 20 for very preterms (<30weeks)

- Saving the sight of one blind child = operating on 10 elderly people with cataract

in terms of “blind person years” saved

Healthy eyes

Regular eye checks, from day one of life

Immunization

Healthy diet

Antenatal care

Face washing